

Markscheme

May 2025

Mathematics: analysis and approaches

Higher level

Paper 3

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Instructions to Examiners

Abbreviations

- M** Marks awarded for attempting to use a correct **Method**.
- A** Marks awarded for an **Answer** or for **Accuracy**; often dependent on preceding **M** marks.
- R** Marks awarded for clear **Reasoning**.
- AG** Answer given in the question and so no marks are awarded.
- FT** Follow through. The practice of awarding marks, despite candidate errors in previous parts, for their correct methods/answers using incorrect results.

Using the markscheme

1 General

Award marks using the annotations as noted in the markscheme *eg M1, A2*.

2 Method and Answer/Accuracy marks

- Do **not** automatically award full marks for a correct answer; all working **must** be checked, and marks awarded according to the markscheme.
- It is generally not possible to award **M0** followed by **A1**, as **A** mark(s) depend on the preceding **M** mark(s), if any.
- Where **M** and **A** marks are noted on the same line, *e.g. M1A1*, this usually means **M1** for an **attempt** to use an appropriate method (*e.g.* substitution into a formula) and **A1** for using the **correct** values.
- Where there are two or more **A** marks on the same line, they may be awarded independently; so if the first value is incorrect, but the next two are correct, award **A0A1A1**.
- Where the markscheme specifies **A3**, **M2** *etc.*, do **not** split the marks, unless there is a note.
- The response to a “show that” question does not need to restate the **AG** line, unless a **Note** makes this explicit in the markscheme.
- Once a correct answer to a question or part question is seen, ignore further working even if this working is incorrect and/or suggests a misunderstanding of the question. This will encourage a uniform approach to marking, with less examiner discretion. Although some candidates may be advantaged for that specific question item, it is likely that these candidates will lose marks elsewhere too.
- An exception to the previous rule is when an incorrect answer from further working is used **in a subsequent part**. For example, when a correct exact value is followed by an incorrect decimal approximation in the first part and this approximation is then used in the second part. In this situation, award **FT** marks as appropriate but do not award the final **A1** in the first part.

Examples:

	Correct answer seen	Further working seen	Any FT issues?	Action
1.	$8\sqrt{2}$	5.65685... (incorrect decimal value)	No. Last part in question.	Award A1 for the final mark (condone the incorrect further working)
2.	$\frac{35}{72}$	0.468111... (incorrect decimal value)	Yes. Value is used in subsequent parts.	Award A0 for the final mark (and full FT is available in subsequent parts)

3 Implied marks

Implied marks appear in **brackets e.g. (M1)**, and can only be awarded if **correct** work is seen or implied by subsequent working/answer.

4 Follow through marks (only applied after an error is made)

Follow through (**FT**) marks are awarded where an incorrect answer from one **part** of a question is used correctly in **subsequent** part(s) (e.g. incorrect value from part (a) used in part (d) or incorrect value from part (c)(i) used in part (c)(ii)). Usually, to award **FT** marks, **there must be working present** and not just a final answer based on an incorrect answer to a previous part. However, if all the marks awarded in a subsequent part are for the answer or are implied, then **FT** marks should be awarded for *their* correct answer, even when working is not present.

For example: following an incorrect answer to part (a) that is used in subsequent parts, where the markscheme for the subsequent part is **(M1)A1**, it is possible to award full marks for *their* correct answer, **without working being seen**. For longer questions where all but the answer marks are implied this rule applies but may be overwritten by a **Note** in the Markscheme.

- Within a question part, once an **error** is made, no further **A** marks can be awarded for work which uses the error, but **M** marks may be awarded if appropriate.
- If the question becomes much simpler because of an error then use discretion to award fewer **FT** marks, by reflecting on what each mark is for and how that maps to the simplified version.
- If the error leads to an inappropriate value (e.g. probability greater than 1, $\sin \theta = 1.5$, non-integer value where integer required), do not award the mark(s) for the final answer(s).
- The markscheme may use the word “their” in a description, to indicate that candidates may be using an incorrect value.
- If the candidate’s answer to the initial question clearly contradicts information given in the question, it is not appropriate to award any **FT** marks in the subsequent parts. This includes when candidates fail to complete a “show that” question correctly, and then in subsequent parts use their incorrect answer rather than the given value.
- Exceptions to these **FT** rules will be explicitly noted on the markscheme.
- If a candidate makes an error in one part but gets the correct answer(s) to subsequent part(s), award marks as appropriate, unless the command term was “Hence”.

5 Mis-read

If a candidate incorrectly copies values or information from the question, this is a mis-read (**MR**). A candidate should be penalized only once for a particular misread. Use the **MR** stamp to indicate that this has been a misread and do not award the first mark, even if this is an **M** mark, but award all others as appropriate.

- If the question becomes much simpler because of the **MR**, then use discretion to award fewer marks.
- If the **MR** leads to an inappropriate value (e.g. probability greater than 1, $\sin \theta = 1.5$, non-integer value where integer required), do not award the mark(s) for the final answer(s).
- Miscopying of candidates' own work does **not** constitute a misread, it is an error.
- If a candidate uses a correct answer, to a "show that" question, to a higher degree of accuracy than given in the question, this is NOT a misread and full marks may be scored in the subsequent part.
- **MR** can only be applied when work is seen. For calculator questions with no working and incorrect answers, examiners should **not** infer that values were read incorrectly.

6 Alternative methods

Candidates will sometimes use methods other than those in the markscheme. Unless the question specifies a method, other correct methods should be marked in line with the markscheme. If the command term is 'Hence' and not 'Hence or otherwise' then alternative methods are not permitted unless covered by a note in the mark scheme.

- Alternative methods for complete questions are indicated by **METHOD 1**, **METHOD 2**, etc.
- Alternative solutions for parts of questions are indicated by **EITHER . . . OR**.

7 Alternative forms

Unless the question specifies otherwise, **accept** equivalent forms.

- As this is an international examination, accept all alternative forms of **notation** for example 1.9 and 1,9 or 1000 and 1,000 and 1.000.
- Do not accept final answers written using calculator notation. However, **M** marks and intermediate **A** marks can be scored, when presented using calculator notation, provided the evidence clearly reflects the demand of the mark.
- In the markscheme, equivalent **numerical** and **algebraic** forms will generally be written in brackets immediately following the answer.
- In the markscheme, some **equivalent** answers will generally appear in brackets. Not all equivalent notations/answers/methods will be presented in the markscheme and examiners are asked to apply appropriate discretion to judge if the candidate work is equivalent.

8 Format and accuracy of answers

If the level of accuracy is specified in the question, a mark will be linked to giving the answer to the required accuracy. If the level of accuracy is not stated in the question, the general rule applies to final answers: *unless otherwise stated in the question all numerical answers must be given exactly or correct to three significant figures*.

Where values are used in subsequent parts, the markscheme will generally use the exact value, however candidates may also use the correct answer to 3 sf in subsequent parts. The markscheme will often explicitly include the subsequent values that come “*from the use of 3 sf values*”.

Simplification of final answers: Candidates are advised to give final answers using good mathematical form. In general, for an **A** mark to be awarded, arithmetic should be completed, and any

values that lead to integers should be simplified; for example, $\sqrt{\frac{25}{4}}$ should be written as $\frac{5}{2}$. An exception to this is simplifying fractions, where lowest form is not required (although the numerator and the denominator must be integers); for example, $\frac{10}{4}$ may be left in this form or written as $\frac{5}{2}$.

However, $\frac{10}{5}$ should be written as 2, as it simplifies to an integer.

Algebraic expressions should be simplified by completing any operations such as addition and multiplication, e.g. $4e^{2x} \times e^{3x}$ should be simplified to $4e^{5x}$, and $4e^{2x} \times e^{3x} - e^{4x} \times e^x$ should be simplified to $3e^{5x}$. Unless specified in the question, expressions do not need to be factorized, nor do factorized expressions need to be expanded, so $x(x+1)$ and $x^2 + x$ are both acceptable.

Please note: intermediate **A** marks do NOT need to be simplified.

9 Calculators

A GDC is required for this paper, but if you see work that suggests a candidate has used any calculator not approved for IB DP examinations (eg CAS enabled devices), please follow the procedures for malpractice.

10 Presentation of candidate work

Crossed out work: If a candidate has drawn a line through work on their examination script, or in some other way crossed out their work, do not award any marks for that work unless an explicit note from the candidate indicates that they would like the work to be marked.

More than one solution: Where a candidate offers two or more different answers to the same question, an examiner should only mark the first response unless the candidate indicates otherwise. If the layout of the responses makes it difficult to judge, examiners should apply appropriate discretion to judge which is “first”.

1. (a) (i) attempts to find an expression for $f^2(x)$ (M1)

Note: For example, award **(M1)** for $f(mx + c)$ seen.

$$= m(mx + c) + c$$

A1

$$= m^2x + cm + c$$

A1

$$f^2(x) = m^2x + c(1 + m)$$

AG

[3 marks]

- (ii) attempts to find an expression for $f^3(x)$ (M1)

Note: For example, award **(M1)** for $f^2(mx + c)$ or $f(m^2x + c(1 + m))$ seen.

EITHER

$$f^3(x) = m^2(mx + c) + c(1 + m) \quad (= m^3x + m^2c + c + cm)$$

A1

OR

$$f^3(x) = m(m^2x + c(1 + m)) + c \quad (= m^3x + mc + m^2c + c)$$

A1

THEN

$$f^3(x) = m^3x + c(1 + m + m^2)$$

AG

[2 marks]

continued...

Question 1 continued

(b) (i) $f^4(x) = m^4x + c(1 + m + m^2 + m^3)$ **A1**

Note: Award **A1** for equivalent forms, for example:

$$f^4(x) = m^4x + c \sum_{i=0}^3 m^i$$

$$f^4(x) = m^3(mx + c) + c(1 + m + m^2)$$

$$f^4(x) = m(m^3x + m^2c + mc + c) + c$$

[1 mark]

(ii) $f^n(x) = m^n x + c(1 + m + (m^2 + \dots + m^{n-1}))$ **A1A1**

Note: Award **A1** for $m^n x$ and **A1** for $c(1 + m + m^2 + \dots + m^{n-1})$ or equivalent,

for example: $c \sum_{i=0}^{n-1} m^i$ or $c \left(\frac{1 - m^n}{1 - m} \right)$

For alternative forms, award **A1** for the correct term in x and **A1** for the correct constant term.

[2 marks]

(iii) **METHOD 1**

attempts to substitute $m = 1$ into their $f^n(x)$ **(M1)**

$$f^n(x) = x + c(1 + 1 + 1 + \dots + 1) \quad \text{A1}$$

$$f^n(x) = x + nc \quad \text{A1}$$

METHOD 2

$$f(x) = x + c \quad \text{A1}$$

attempts to find $f^2(x)$ with their $f(x)$ and $m = 1$ **(M1)**

$$f^2(x) = f(x + c) = x + 2c$$

$$\Rightarrow f^n(x) = x + nc \quad \text{A1}$$

[3marks]

continued...

Question 1 continued

(c) consider $n = 1$

$$f^1(x) = m^1x + c \left(\frac{1-m^1}{1-m} \right) \Rightarrow f(x) = mx + c,$$

R1

hence true for $n = 1$

Note: The substitution $n = 1$ must be explicit for the **R1** to be awarded.

Award **R0** for only considering $n = 2$.

Subsequent marks after this **R1** mark are independent of this mark and can be awarded.

assume true for $n = k$ ($p(k)$ is true)

M1

$$f^k(x) = m^kx + c \left(\frac{1-m^k}{1-m} \right) \text{ (for some } k \in \mathbb{Z}^+ \text{)}$$

Note: The assumption of truth must be apparent. Do not award **M1** for statements such as “let $n = k$ ” or “ $n = k$ is true” or “assume $n = k$ is true”. Subsequent marks after this **M1** are independent of this mark and can be awarded.

EITHER

Attempting to find $f(f^k(x))$

(M1)

$$f^{k+1}(x) = f(f^k(x))$$

$$f\left(m^kx + c \left(\frac{1-m^k}{1-m} \right)\right) = m\left(m^kx + c \left(\frac{1-m^k}{1-m} \right)\right) + c$$

A1

$$= \left(m^{k+1}x + cm \left(\frac{1-m^k}{1-m} \right) \right) + c$$

OR

Attempting to find $f^k(f(x))$

(M1)

$$f^{k+1}(x) = f^k(f(x))$$

$$f^k(mx + c) = m^k(mx + c) + c \left(\frac{1-m^k}{1-m} \right)$$

A1

$$= \left(m^{k+1}x + cm^k + c \left(\frac{1-m^k}{1-m} \right) \right)$$

continued...

Question 1 continued

THEN

EITHER

attempts to put c over a common denominator **(M1)**

$$m^{k+1}x + cm \left(\frac{1-m^k}{1-m} \right) + \frac{c(1-m)}{1-m}$$

$$= m^{k+1}x + c \left(\frac{m(1-m^k) + (1-m)}{1-m} \right) \text{ (or equivalent)} \quad \textbf{A1}$$

OR

attempts to put cm^k over a common denominator **(M1)**

$$m^{k+1}x + \frac{cm^k(1-m)}{1-m} + c \left(\frac{1-m^k}{1-m} \right)$$

$$= m^{k+1}x + c \left(\frac{m^k(1-m) + (1-m^k)}{1-m} \right) \text{ (or equivalent)} \quad \textbf{A1}$$

OR

recognizes that $\frac{1-m^k}{1-m}$ can be expressed as a geometric series **(M1)**

$$m^{k+1}x + cm(1 + m + m^2 + \dots + m^{k-1}) + c$$

$$= m^{k+1}x + c(1 + m + m^2 + m^3 + \dots + m^k) \quad \textbf{A1}$$

THEN

$$= m^{k+1}x + c \left(\frac{1-m^{k+1}}{1-m} \right) \quad \textbf{A1}$$

since true for $n = 1$ and true for $n = k + 1$ if true for $n = k$, hence true for all $n (\in \mathbb{Z}^+)$ **R1**

Note: To obtain the final **R1**, four of the previous seven marks must have been awarded.

The statement ‘true for $n = 1$ ’ may be seen anywhere in the proof.

[8 marks]

continued...

Question 1 continued

- (d) attempts to apply $m^n \rightarrow 0$ (as $n \rightarrow \infty$) to at least one term in expression (M1)

Note: Award (M1) for an equivalent statement/conclusion in words.

$$m^n x \rightarrow 0 \qquad \left(c \left(\frac{1-m^n}{1-m} \right) \rightarrow \right) \frac{c}{1-m} \text{ (as } n \rightarrow \infty \text{)} \qquad \text{(A1)A1}$$

Note: Award (A1) for the correct term in x and A1 for the correct constant term.

$$y = \frac{c}{1-m} \qquad \text{A1}$$

[4 marks]

(e) (i) $f^n(x) = (-1)^n x + c \left(\frac{1-(-1)^n}{1-(-1)} \right)$ A1

Note: Award A1 here if the above is seen in part (e)(ii).

Award A0 for a missing bracket, for example: $f^n(x) = -1^n x + c \left(\frac{1-(-1)^n}{1-(-1)} \right)$.

$$(-1)^n = -1 \text{ (when } n \text{ is odd)} \qquad \text{R1}$$

$$\left(f^n(x) = -x + c \left(\frac{1-(-1)}{1-(-1)} \right) \right)$$

Note: The A1 and R1 mark can be awarded in either order. A0R1 is possible.

Award R0 for arguments based on specific numerical examples.

$$f^n(x) = -x + c \qquad \text{AG}$$

[2 marks]

- (ii) evidence that an even n has been considered (M1)

$$\text{for example, } (-1)^2 = 1$$

$$f^n(x) = x \qquad \text{A1}$$

Note: Award (M1)A1 for answer of x (need not be an equation).

[2 marks]

Total [27 marks]

2. (a) (i) 0

A1
[1 mark]

(ii) 0

A1
[1 mark]

(b) attempts to expand a product of two of the side lengths

(M1)

EITHER

$$V = x(4x^2 - 2(a+b)x + ab) \quad (V = x(4x^2 - 2ax - 2bx + ab))$$

A1

OR

$$V = (ax - 2x^2)(b - 2x)$$

$$= 4x^3 - 2ax^2 - 2bx^2 + abx$$

A1

OR

$$V = (a - 2x)(bx - 2x^2)$$

$$= 4x^3 - 2ax^2 - 2bx^2 + abx$$

A1

THEN

$$V = 4x^3 - 2(a+b)x^2 + abx$$

AG
[2 marks]

continued...

Question 2 continued

- (c) attempts to find $\frac{dV}{dx}$ (M1)

$$\frac{dV}{dx} = 12x^2 - 4(a+b)x + ab \quad \text{A1}$$

Note: Award (M1) for obtaining the equivalent of two correct terms.

EITHER

- attempts to solve $\frac{dV}{dx} = 0$ for x using the quadratic formula (M1)

$$x = \frac{-(-4(a+b)) \pm \sqrt{(-4(a+b))^2 - 4(12)(ab)}}{24} \quad \text{(A1)}$$

Note: Award A0 for incorrect expressions, for example:

$$x = \frac{-(-4(a+b)) \pm \sqrt{-4(a+b)^2 - 4(12)(ab)}}{24}$$

- attempts to expand brackets within the square root sign and/or to take a common factor of 4 (M1)

$$x = \frac{-(-4(a+b)) \pm \sqrt{16a^2 + 16b^2 + 32ab - 48ab}}{24} \quad \text{A1}$$

or $x = \frac{4(a+b) \pm 4\sqrt{a^2 + 2ab + b^2 - 3ab}}{24}$

OR

- attempt to substitute either of the given x values into their $\frac{dV}{dx}$ M1

- correct manipulation with each solution to reduce each expression to zero A1A1

- $\frac{dV}{dx} = 0$ is a quadratic equation and hence these are the only two solutions R1

THEN

$$x = \frac{(a+b) \pm \sqrt{a^2 - ab + b^2}}{6} \quad \text{AG}$$

[6 marks]

continued...

Question 2 continued

- (d) attempts to find their $\frac{d^2V}{dx^2}$ (M1)

$$\frac{d^2V}{dx^2} = 24x - 4(a + b) \quad \text{A1}$$

$$\frac{d^2V}{dx^2} < 0 \quad \text{R1}$$

Note: Award **R1** for equivalent statement in words.

substitutes $x = \frac{(a+b) - \sqrt{a^2 - ab + b^2}}{6}$ into their $\frac{d^2V}{dx^2}$ (M1)

Note: Award (M1) for substituting $x = \frac{(a+b) \pm \sqrt{a^2 - ab + b^2}}{6}$

$$\begin{aligned} \frac{d^2V}{dx^2} &= 24 \left(\frac{(a+b) - \sqrt{a^2 - ab + b^2}}{6} \right) - 4(a+b) \\ &= -4\sqrt{a^2 - ab + b^2} (< 0) \quad \text{A1} \end{aligned}$$

so $x_m = \frac{(a+b) - \sqrt{a^2 - ab + b^2}}{6}$ gives a local maximum of V AG

[5 marks]

- (e) the maximum volume will occur at either the endpoints or at the local maximum. R1

Note: Award **R1** for an answer which refers to the volume being 0 at both endpoints ($x = 0$ and $x = \frac{a}{2}$) and the volume of the box at $x_m > 0$

hence the maximum volume of the box occurs at x_m / the local maximum is a global maximum.

AG
[1 mark]

continued...

Question 2 continued

(f) $2st - s^2 < t^2 - s^2$ A1

$t^2 > 2st$ and since $t \in \mathbb{Z}^+$ (dividing both sides by t) R1

Note: Award **R1** for equivalent reasoning, $t > 0$ or t is positive

$\Rightarrow t > 2s$ AG
[2 marks]

(g) attempts to substitute for each of a, b and $\sqrt{a^2 - ab + b^2}$ in terms of s and t (M1)

$(x_m =) \frac{(2st - s^2) + (t^2 - s^2) - (s^2 - st + t^2)}{6}$ A1

$= \frac{-3s^2 + 3st}{6}$ A1

$x_m = \frac{s(t-s)}{2}$ AG

[3 marks]

(h) (i) **METHOD 1**

when s is even, x_m is an integer R1

hence V_m is a product of integers ($a, b \in \mathbb{Z}^+$) R1

hence V_m is an integer AG

METHOD 2

$V_m = \frac{s(t-s)}{2} \left(2st - s^2 - 2 \left(\frac{s(t-s)}{2} \right) \right) \left(t^2 - s^2 - 2 \left(\frac{s(t-s)}{2} \right) \right)$ (or equivalent) A1

Note: Award **A1** for $V_m = \frac{s(t-s)}{2} \left(a - 2 \left(\frac{s(t-s)}{2} \right) \right) \left(b - 2 \left(\frac{s(t-s)}{2} \right) \right)$

when s is even, $\frac{s(t-s)}{2}$ is an integer **AND**

hence V_m is a product of integers ($a, b \in \mathbb{Z}^+$) R1

hence V_m is an integer AG

[2 marks]

continued...

Question 2 continued

(ii) when s is odd **AND** t is odd ($s(t-s)$ is even)

A1

x_m and V_m are both integers

AG

[1 mark]

(iii) the following table shows some possible sets of values for s, t, a, b, x_m and V_m

s	t	a	b	x_m	V_m
1	3	5	8	1	18
1	5	9	24	2	200
2	5	16	21	3	450
2	6	20	32	4	1152
2	7	24	45	5	2450
2	8	28	60	6	4608
3	7	33	40	6	3528
4	10	64	84	12	28800
...	...	...	...	...	...

attempts to use values of s and t to find

EITHER x_m **OR** a and b **OR** V_m

(M1)

a corresponding correct x_m value

A1

corresponding correct a and b values

(A1)

a corresponding correct V_m value

A1

Note: Award at most **(M1)A0A0A0** if the values of s and t do not satisfy the conditions: $s, t \in \mathbb{Z}^+$, $t > 2s$, and s is even / s and t are odd.

The three **A** marks can be awarded in any order.

The **A1** for x_m and the **A1** for a and b can be awarded independently.

Some candidates may determine in part (h) that $V_m = \frac{s^2 t^2 (t-s)^2}{2}$.

[4 marks]

Total [28 marks]